AdanPradan / FairShare — Platform Functional & Backend Specification

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0. Executive Summary

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AdanPradan (consumer brand) / FairShare (B2B SaaS brand) is a plugin-first, multitenant marketplace platform connecting vendors, customers, delivery partners, and the operator through six portals. Each portal has its own auth scope, data visibility rules, and workflow ownership.

Portal	Primary User	Core Job
Platform Admin	Ninepoint (operator)	Govern the platform: approve plugins, vendors, products; manage all users and tenants
Tenant / Org Admin	Tenant or Org admin	Manage members, settings, billing, vendor relationships within their scope
Vendor Portal	Vendor staff	List products, upload images, fulfil orders, view payouts
Customer Marketplace	End consumer	Browse, search, cart, checkout, track orders, raise returns
Delivery Partner Portal	Delivery agent	Accept pickups, navigate, mark delivery, upload proof
Chargeback Portal	Platform admin + vendor	Dispute resolution: evidence upload, adjudication, refund trigger

Key architectural principles:

- Every vendor product submission passes a Platform Admin content review gate before going live to customers.
- Products are visible only in cities the vendor has explicitly enabled.
- Standard Plugins are published by the operator and available to all verified vendors immediately; Vendor Plugins require PA review.

- The first user created on the platform is auto-assigned Platform Admin; all others start as Individual tenants.
- All financial flows (payouts, refunds, chargebacks) are mediated by the platform; vendors never hold funds directly.

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1. Platform Admin Portal

1.1 Platform Team Management

A single operator company runs the platform, but their internal team can be large — KYC reviewers, content moderators, finance staff, support agents. Each person gets a platform-level role that scopes exactly what they can see and do. No platform team member belongs to a tenant; they all exist above the tenant layer and are managed entirely within the Platform Admin portal.

1.1.1 Platform Role Hierarchy

Role	Who Holds It	Scope
Super Admin	First registered user on the platform (auto-promoted); exactly one per platform	Full access; cannot be demoted; can transfer designation to another PA
Platform Admin	Invited by Super Admin or another Platform Admin	Full access; can invite and manage team members
Finance Manager	Invited by PA	Chargebacks, refunds, vendor payouts and holds only
Content Moderator	Invited by PA	Product Content Review queue only; no access to financials or plugins
KYC Reviewer	Invited by PA	Vendor onboarding applications and document review only
Support Agent	Invited by PA	Read-only view of orders, customers, tenants; can raise internal tickets; cannot approve or reject anything

Role permissions are additive downward. A Platform Admin implicitly holds every permission of every role beneath it. Super Admin adds the ability to create service accounts, transfer the Super Admin designation, and modify platform-level config keys.

1.1.2 Role-Scoped Sidebar Visibility

Sidebar Section	Super Admin	Platform Admin	Finance Manager	Content Moderator	KYC Reviewer	Support Agent
Dashboard	Full	Full	Finance KPIs only	Review KPIs only	KYC KPIs only	Order KPIs only
Plugin Review	Yes	Yes	No	No	No	No
Content Review Queue	Yes	Yes	No	Yes	No	No
Vendor Onboarding	Yes	Yes	No	No	Yes	Read-only
User Directory	Yes	Yes	No	No	No	Read-only
Vendor Directory	Yes	Yes	No	No	Yes	Read-only
Tenant Management	Yes	Yes	No	No	No	Read-only
Chargebacks and Refunds	Yes	Yes	Yes	No	No	No
Payouts	Yes	Yes	Yes	No	No	No
Analytics	Yes	Yes	Finance only	No	No	No
Audit Log	Yes	Yes	Own actions only	Own actions only	Own actions only	Own actions only
Platform Config	Yes	Yes	No	No	No	No
Team Management	Yes	Yes	No	No	No	No

Sidebar sections are rendered server-side based on the JWT role claim. A missing section never returns a 404 for a wrong-role user — it simply does not appear in the navigation. Direct URL access to a restricted section returns 403 and redirects to the role's default landing page.

1.1.3 Team Member Invite Flow

1. Super Admin or Platform Admin opens Team Management in the sidebar.
2. Clicks Invite Team Member button.
3. Enters the invitee's email address.
4. Selects a platform role from the dropdown (all roles except Super Admin are selectable).

5. Optionally adds a personal note that appears in the invite email.
6. Clicks Send Invite.
7. Invitee receives a secure join link (valid 48 hours, one-time use, stored as a hash).
8. Invitee clicks the link, sets a password, and completes mandatory MFA setup (TOTP via authenticator app).
9. Account is activated; invitee appears in the Team list with status Active.

If the link expires before use, any Platform Admin can resend the invite from the Invited row in the team table. A pending invite can also be revoked before acceptance. The invitee's email cannot be changed after an invite is sent; a new invite must be issued for a different address.

1.1.4 Team Member States

State	Meaning	Available Actions
Invited	Email sent; join link not yet used	Resend invite, Revoke invite
Active	MFA set up; member can log in	Edit role, Suspend, Reset MFA
Suspended	Access blocked; sessions invalidated; data and audit log intact	Reactivate, Delete

Suspending a team member immediately invalidates all their active access tokens and refresh tokens. Their historical audit log entries are preserved indefinitely.

1.1.5 Role Change Rules

A Platform Admin can change the role of any team member except the Super Admin. Promotion to Super Admin is not available through the role change UI — it requires the current Super Admin to initiate a designation transfer, which demotes themselves to Platform Admin and promotes the selected PA. This transfer requires the Super Admin to re-enter their password as confirmation. Every role change is written to the audit log with actor, old role, new role, and timestamp. Role changes take effect immediately; the affected member's next API call is re-evaluated against the new role's permissions.

1.1.6 Key Data Entities

Entity	Fields
PlatformTeamMember	id, email, name, platform_role, status (invited/active/suspended), invited_by, invited_at, activated_at, last_active_at, mfa_enabled
PlatformInvite	id, email, platform_role, token_hash, invited_by, invited_at, expires_at, accepted_at, revoked_at

1.1.7 API Endpoints

Method	Path	Description
GET	/api/pa/team	List all platform team members (filter: role, status)
POST	/api/pa/team/invite	Invite a new team member (email + role)
PATCH	/api/pa/team/:id/role	Change a team member's platform role
POST	/api/pa/team/:id/suspend	Suspend a team member (invalidates all sessions)
POST	/api/pa/team/:id/reactivate	Reactivate a suspended team member
POST	/api/pa/team/:id/reset-mfa	Force MFA reset on the member's next login
DELETE	/api/pa/team/invites/:id	Revoke a pending invite
POST	/api/pa/team/invites/:id/resend	Resend an expired or pending invite
POST	/api/pa/team/super-admin/transfer	Transfer Super Admin designation (Super Admin only)

1.1.8 Functional Test Cases

TC	Scenario	Steps	Expected Result
PT-01	First user auto-becomes Super Admin	Register the first user on a fresh platform instance	Role = Super Admin; Team Management visible; no invite required

TC	Scenario	Steps	Expected Result
PT-02	Invite a Finance Manager	PA enters email, selects Finance Manager, clicks Send Invite	Invite email sent with 48-hour join link; member listed as Invited
PT-03	Invitee completes onboarding	Invitee clicks link within 48h, sets password, scans TOTP QR code	Account activated; status = Active; MFA = enabled
PT-04	Invite link expires	Invitee clicks link after 48 hours	Error: invite link expired; PA can resend from team table
PT-05	Role-scoped sidebar for Content Moderator	Log in as Content Moderator	Sidebar shows only Dashboard (review KPIs) and Content Review Queue; all other sections absent
PT-06	Finance Manager blocked from plugin review	Finance Manager navigates directly to /pa/plugins URL	403 returned; redirect to Finance Dashboard
PT-07	Role change takes effect immediately	PA changes Support Agent to KYC Reviewer	Next API call from that user returns KYC Reviewer permissions; old Support Agent routes return 403
PT-08	Suspend invalidates active session	PA suspends a currently logged-in team member	Member's next API call returns 401; login page shows suspended account message
PT-09	Super Admin transfer	Super Admin initiates transfer to a Platform Admin	Transferor becomes Platform Admin; transferee becomes Super Admin; both notified by email; audit log records actor, old role, new role, timestamp
PT-10	PA cannot demote Super Admin	PA attempts to change Super Admin's role via role change UI	Role dropdown for Super Admin is disabled; action blocked; error shown

1. Platform Admin Portal

The Platform Admin portal is the operator control plane. Only service accounts and the first-created user (auto-promoted to Platform Admin role) can access it. All governance actions — vendor approval, plugin review, product content moderation, user suspension — originate here.

1.1 Access & Roles

Role	How Assigned	Scope
Platform Admin	Auto (first user) or manual grant	Full platform
Service Account	Created by PA only	API-key auth, scoped permissions

MFA is mandatory for all Platform Admin accounts. Sessions expire after 8 hours.

1.2 Feature Modules

Plugin Review

- View all vendor-submitted plugins with status: Pending / In Review / Approved / Rejected / Published.
- Filter by risk level (Low / Medium / High), category, vendor.
- Open a plugin submission to review: description, capabilities claimed, permissions requested, sandbox test results.
- Actions: Approve → status becomes Published; Reject → vendor notified with reason; Request Changes → vendor can resubmit.
- Risk scoring is auto-calculated from capabilities claimed (payments, webhooks, OAuth2 = higher risk).

Standard Plugin Management

- PA creates platform-owned plugins available to all verified vendors without approval.
- Fields: name, version, category (Payments / Tax / Shipping / Auth), capabilities, description.
- Publish / Unpublish / Deprecate actions.

Product Content Review Queue

- All new vendor product submissions land here with status Pending Review.
- Review checklist per listing: images meet spec (min 2, 800x800px square, white background, no watermarks), description matches images, MRP is accurate, no misleading claims, all 5 vendor declarations signed.

- Actions: Approve → product status becomes Live; Reject → vendor notified with reason (pre-set reasons available: images below spec, MRP mismatch, misleading claims, watermark present, too few images).
- Listings with 0 images or fewer than 5 declarations cannot be approved (Approve button disabled).

User Directory

- View all users across all tenants: User Accounts and Service Accounts.
- Filter by role, account type, status.
- Actions: Suspend (blocks access, data intact), Block (permanent ban), Reset MFA, Promote to Tenant Admin.
- Account states: Pending → Active → Suspended / Blocked.

Vendor Directory

- View all vendors: Active, Pending KYC, Suspended.
- Trigger KYC review, suspend a vendor, invite a new vendor.
- View per-vendor: products count, locations covered, plugins installed, GMV MTD.

Tenant Management

- View all tenants: Individual, Organisation, Vendor.
- Upgrade path: Individual → Organisation → Vendor (KYC + GST cert + bank details required for Vendor upgrade).
- Actions: Provision new tenant, force-upgrade tier, suspend tenant.

Roles & Permissions

- Full RBAC matrix: Platform Admin / Tenant Admin / Org Admin / Vendor Admin / Vendor Member / Customer / Viewer.
- Create custom roles (based on an existing role, with permission overrides).
- Assign roles per user per tenant scope.

Products & Locations

- Unified view of all products across vendors with PA Status (Live / Pending Review / Rejected).
- Location Map: per-city vendor and product counts.
- Vendor Coverage Matrix: which vendors serve which cities.
- Relationship Flow: Vendor → Product → Location → Customer dependency diagram.

Analytics & Audit

- GMV trend, top vendors by revenue, top products by orders.

- Full audit log: every PA action timestamped with actor, target, old/new value.
- Platform config: commission rate, plugin sandbox settings, feature flags.

1.3 Key Data Entities

Entity	Fields
Plugin	id, name, vendor_id, version, category, capabilities[], risk_level, status, submitted_at, reviewed_at, reviewer_id, reject_reason
Product	id, vendor_id, name, category, price_mrp, images[], cities[], stock_status, pa_status, submitted_at, approved_at, reject_reason
User	id, email, account_type, role, tenant_id, status, mfa_enabled, last_active_at
Vendor	id, tenant_id, kyc_status, gstin, bank_account_id, gmv_mtd, status
Tenant	id, name, type (Individual/Organisation/Vendor), plan, created_at, status
AuditLog	id, actor_id, action, target_type, target_id, old_value, new_value, timestamp

1.4 API Endpoints

Method	Path	Description
GET	/api/pa/plugins	List plugin submissions (filter: status, risk, vendor)
PATCH	/api/pa/plugins/:id/approve	Approve a plugin
PATCH	/api/pa/plugins/:id/reject	Reject with reason
GET	/api/pa/products/pending	List pending product submissions
PATCH	/api/pa/products/:id/approve	Approve product listing
PATCH	/api/pa/products/:id/reject	Reject with reason
GET	/api/pa/users	List all users (filter: role, type, status)
PATCH	/api/pa/users/:id/suspend	Suspend a user

Method	Path	Description
GET	/api/pa/vendors	List vendors
PATCH	/api/pa/vendors/:id/kyc-approve	Approve vendor KYC
GET	/api/pa/audit-log	Fetch audit log (filter: actor, action, date range)
GET	/api/pa/analytics/gmv	GMV trend data

All PA endpoints require `Authorization: Bearer <pa-token>` and role claim `platform_admin`.

1.5 Functional Test Cases

TC	Scenario	Steps	Expected Result
PA-01	Approve a pending plugin	Log in as PA, open Plugin Review, click Approve on a Pending plugin	Plugin status → Published; vendor receives approval notification
PA-02	Reject a plugin with reason	Click Reject, select reason, submit	Plugin status → Rejected; reason stored; vendor notified
PA-03	Approve a product with valid images and declarations	Open Content Review, select listing with 2+ images and 5/5 declarations, click Approve	Product status → Live; visible to customers in assigned cities
PA-04	Approve button disabled for listing with 0 images	Open listing with 0 images	Approve button is greyed out; tooltip explains reason
PA-05	Reject a product — MRP mismatch	Click Reject, select preset reason MRP mismatch, submit	Vendor receives rejection with reason; product stays Pending
PA-06	Suspend a user	User Directory → find user → Suspend	User cannot log in; data intact; audit log records action

TC	Scenario	Steps	Expected Result
PA-07	First user auto-gets Platform Admin	Create a fresh platform instance, register first user	User role = Platform Admin; no other role option presented
PA-08	Service account cannot log in via password	Attempt password login with a service account email	Login refused with error: use API key auth
PA-09	Audit log records every PA action	Perform 3 PA actions (approve product, suspend user, reject plugin)	All 3 appear in audit log with actor, timestamp, old/new values
PA-10	Custom role with overridden permission	Create custom role based on Tenant Admin with -manage:billing	Users with this role cannot access billing; all other Tenant Admin permissions work

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2. Tenant & Organisation Admin Portal

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A tenant is the top-level account holder on the platform. Every user, vendor, and organisation belongs to exactly one tenant. This portal allows tenant owners (and the organisation admins they delegate to) to manage their internal hierarchy, members, billing, and sub-scoped plugins.

2.1 Tenant Types & Upgrade Path

Type	Who Uses It	Key Unlock
Individual	Solo consumer or freelancer	Marketplace purchasing only
Organisation	Business / team with multiple members	Member management, org-scoped plugins, shared address book
Vendor	Registered seller	Product listings, order fulfilment, revenue dashboard

Upgrade flow: Individual to Organisation (Tenant Admin submits business registration number) to Vendor (KYC + GST certificate + bank account, reviewed by PA). Downgrade is blocked if vendor has live listings or outstanding payouts.

2.2 Tenant Admin Features

Member Management

- Invite members by email; they receive a join link valid for 48 hours.
- Assign roles: Org Admin, Member, Viewer.
- Revoke access: member loses portal access immediately; historical audit records are preserved.
- Member states: Invited, Active, Suspended.

Organisation Profile

- Legal name, registered address, GSTIN (used on invoices), primary contact, logo upload.
- Billing email distinct from login email.

Billing & Subscription

- View current plan (Free, Starter, Growth, Enterprise).
- Upgrade or downgrade plan (proration calculated at change time).
- Payment methods: add/remove credit card or bank account via Razorpay.
- Invoice history: download PDF per invoice.
- Usage meter: API calls, active products, monthly orders vs. plan limits.

Org-Scoped Plugin Management

- Org Admins can install platform-approved plugins scoped to their organisation.
- Plugin marketplace shows available plugins with permissions and pricing.
- Configure installed plugins: credentials, webhook URLs, feature toggles.
- View plugin activity logs.

Shared Address Book

- Org-level delivery addresses shared across all members.
- Add, edit, archive addresses.
- Set default delivery address per member.

SSO / Auth Settings (Growth and Enterprise plans)

- Configure SAML 2.0 or OIDC identity provider.
- Map IdP groups to platform roles.

- Enforce SSO: members cannot use password login once SSO is active.
- Test SSO connection before enforcing.

2.3 Organisation Admin Features

Org Admins are delegated by the Tenant Admin. They cannot change billing or SSO settings but can manage members within their organisation scope, install and configure org-scoped plugins, manage the shared address book, view org-level order and spending reports, and raise support tickets on behalf of the organisation.

2.4 Key Data Entities

Entity	Fields
Tenant	id, name, type, plan, gstin, billing_email, status, upgraded_at
Member	id, tenant_id, user_id, role, status, invited_at, activated_at
OrgPlugin	id, tenant_id, plugin_id, installed_by, config_json, status, installed_at
BillingPlan	id, name, price_monthly, limits{api_calls, products, orders}
Invoice	id, tenant_id, plan_id, amount, period_start, period_end, pdf_url, paid_at
Address	id, tenant_id, label, line1, line2, city, state, pincode, is_default

2.5 API Endpoints

Method	Path	Description
GET	/api/tenant/:id/members	List members (filter: role, status)
POST	/api/tenant/:id/members/invite	Invite a member by email
PATCH	/api/tenant/:id/members/:uid/role	Change member role
DELETE	/api/tenant/:id/members/:uid	Revoke member access
GET	/api/tenant/:id/billing	Fetch current plan, usage, next invoice

Method	Path	Description
POST	/api/tenant/:id/billing/upgrade	Change plan
GET	/api/tenant/:id/plugins	List installed org plugins
POST	/api/tenant/:id/plugins/:pid/install	Install a plugin
PATCH	/api/tenant/:id/plugins/:pid/config	Update plugin configuration
GET	/api/tenant/:id/addresses	List shared addresses
POST	/api/tenant/:id/sso/configure	Configure SSO provider
POST	/api/tenant/:id/sso/test	Test SSO connection

2.6 Functional Test Cases

TC	Scenario	Steps	Expected Result
TA-01	Invite a member	Tenant Admin opens Members, enters email, selects role Member, clicks Invite	Invite email sent; member shows as Invited in list
TA-02	Member activates invite	New member clicks join link within 48 hours	Account created; member status becomes Active
TA-03	Invite link expires	Member clicks link after 48 hours	Error: invite link expired; Tenant Admin must re-invite
TA-04	Revoke member access	Tenant Admin clicks Revoke on an Active member	Member cannot log in; their historical orders visible to TA; audit log updated
TA-05	Upgrade tenant to Organisation	Individual tenant submits business registration number	PA notified; tenant type changes to Organisation on approval
TA-06	Plan upgrade proration	Upgrade from Starter to Growth on day 15 of billing cycle	Invoice shows prorated charge for remaining days; new limits apply immediately

TC	Scenario	Steps	Expected Result
TA-07	Install org plugin	Org Admin opens Plugin Marketplace, installs Razorpay plugin	Plugin appears in Installed list; config panel opens
TA-08	SSO enforce blocks password login	Tenant Admin enables Enforce SSO	Members who try password login are redirected to IdP
TA-09	Org Admin cannot change billing	Log in as Org Admin, navigate to Billing	Billing tab hidden or read-only; no Upgrade button visible
TA-10	Downgrade blocked with live vendor listings	Attempt to downgrade Vendor tenant to Organisation	Error: cannot downgrade as 14 live product listings exist; remove listings first

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3. Vendor Onboarding Process

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Vendor onboarding is a gated, multi-step workflow that ends with a vendor account receiving a Verified badge and the ability to publish live product listings. The Platform Admin is the final approver at each gate.

3.1 Onboarding Stages

Stage	Actor	Action	Outcome
1. Register	Applicant	Creates individual account	Tenant created (type: Individual)
2. Express Intent	Applicant	Clicks Become a Vendor	Vendor application form unlocked
3. Business Profile	Applicant	Fills legal name, address, category, description	Draft application saved
4. Document Upload	Applicant	Uploads GST certificate, PAN, bank statement	Documents attached to application

Stage	Actor	Action	Outcome
5. Bank Account	Applicant	Enters account number, IFSC; Razorpay penny-drop verification	Bank account verified or failed
6. Submit	Applicant	Clicks Submit for Review	Application status: Pending KYC
7. PA Review	Platform Admin	Opens application, checks documents and bank account	Approve (status: Verified) or Reject (with reason)
8. Onboarding Complete	Vendor	Receives approval email	Tenant upgraded to Vendor; portal unlocked

3.2 Application Form Fields

Business Profile: Legal business name (as on GST certificate), business type (Sole Proprietorship / Partnership / Private Limited / LLP / Other), registered address, primary category (Grocery / Dairy / Beverages / Snacks / Household / Other), business description (max 500 characters), primary contact name, phone, email.

Documents Required: GST Certificate (PDF, max 5 MB), PAN Card of business or proprietor (PDF or image, max 5 MB), cancelled cheque or bank statement not older than 3 months (PDF, max 5 MB), FSSAI licence (optional at application stage, mandatory for food vendors before first product listing).

Bank Account Details: Account holder name, account number, IFSC code, bank name and branch (auto-filled from IFSC lookup), penny-drop verification (platform transfers Re 1; vendor confirms the transaction reference from their bank statement).

3.3 PA Review Checklist

When a PA opens a Pending KYC application they see a document viewer (inline PDF/image preview), GST number validity check (API call to GST portal returns Active / Inactive / Not Found), bank penny-drop result, business name match between GST certificate and form, and internal review notes.

Actions available: Approve, Reject (with reason), Request Resubmission (vendor can upload corrected documents without losing the application). Common preset reject reasons: GST certificate expired or inactive, PAN name does not match business name, bank account penny-drop failed, documents illegible or incomplete, FSSAI licence required for food category.

3.4 Post-Approval Steps

Once approved, the vendor completes a brief setup wizard: set operating cities (min 1, max 20 at launch), set operating hours per city, upload brand logo and banner image, configure delivery (self-delivery or platform delivery partners), list first product with inline guidance. Vendors who complete the wizard move to Active status. Vendors who skip it remain in Setup Pending and cannot receive orders.

3.5 Key Data Entities

Entity	Fields
VendorApplication	id, tenant_id, status, business_name, business_type, gstin, pan, category, submitted_at, reviewed_at, reviewer_id, reject_reason, resubmit_count
Document	id, application_id, type (gst/pan/bank/fssai), file_url, uploaded_at, verified
BankVerification	id, application_id, account_number, ifsc, holder_name, penny_drop_status, verified_at

3.6 API Endpoints

Method	Path	Description
POST	/api/onboarding/apply	Start a vendor application
PATCH	/api/onboarding/:id/profile	Save business profile fields
POST	/api/onboarding/:id/documents	Upload a document
POST	/api/onboarding/:id/bank/initiate	Start penny-drop verification
PATCH	/api/onboarding/:id/bank/confirm	Confirm penny-drop reference
POST	/api/onboarding/:id/submit	Submit application for PA review
GET	/api/pa/onboarding	List all applications (filter: status)
PATCH	/api/pa/onboarding/:id/approve	Approve application
PATCH	/api/pa/onboarding/:id/reject	Reject with reason
PATCH	/api/pa/onboarding/:id/request-resubmission	Ask vendor to resubmit documents

3.7 Functional Test Cases

TC	Scenario	Steps	Expected Result
VO-01	Happy path full onboarding	Register, fill all fields, upload valid documents, complete penny-drop, submit	Application reaches PA queue with status Pending KYC
VO-02	GST API returns inactive	PA opens application; GST check runs	Red warning badge on GST document; PA sees Inactive status
VO-03	Bank penny-drop fails	Vendor enters wrong account number	Penny-drop fails; vendor shown error with option to re-enter bank details
VO-04	Request resubmission	PA clicks Request Resubmission with note about illegible PAN	Vendor receives email with note; can upload new PAN without resetting application

TC	Scenario	Steps	Expected Result
VO-05	Reject application	PA clicks Reject, selects preset reason GST certificate expired	Vendor receives rejection email with reason; application closed
VO-06	Approve application	PA clicks Approve	Vendor receives approval email; tenant type becomes Vendor; portal unlocked
VO-07	Vendor skips setup wizard	Approved vendor logs in and dismisses setup wizard	Status remains Setup Pending; vendor cannot publish products or receive orders
VO-08	Food vendor without FSSAI	Vendor in Grocery category tries to list first product	Warning: FSSAI licence required before first product listing
VO-09	Duplicate GSTIN	Second application submitted with same GSTIN as existing vendor	Submission blocked with error: GSTIN already registered
VO-10	Resubmit count tracked	Vendor resubmits documents twice	PA sees resubmit count = 2 on application; can flag for manual escalation

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4. Vendor Portal

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The Vendor Portal is the day-to-day operating dashboard for a verified vendor. It covers product catalogue management (with the PA content review gate), order fulfilment, earnings, locations, plugins, and compliance declarations.

4.1 Dashboard

Summary cards show Today's Orders, Revenue MTD, Active Products, Pending Reviews, Rejected Listings, and Cities Covered. Alert banners prompt the vendor if they have listings awaiting PA review, rejected listings needing correction, or expiring compliance documents. Quick actions cover Add Product, View Orders, and Download Earnings Report.

4.2 Product Catalogue

Adding a New Product: Fields include product name, category, sub-category, description (rich text, max 2000 characters), MRP, selling price, unit (grams / ml / pieces / kg / litre), SKU code, and stock status (In Stock / Out of Stock / Coming Soon).

Images: minimum 2, maximum 10 per listing. Each image must be at least 800x800 px, square crop, white background, no watermarks. Drag-and-drop upload with inline pixel-dimension validation.

Compliance Declarations (all 5 must be checked before submission): (1) Product images are authentic and not stock photos. (2) MRP is correct as per product packaging. (3) Product is legal to sell in all listed cities. (4) No misleading health, medical or nutritional claims. (5) Vendor holds all required licences (FSSAI, etc.) for this product.

On Submit, product status becomes Pending Review and is invisible to customers until PA approves.

Editing a Live Product creates a new pending revision. The live version remains visible to customers until the revision is approved.

Withdrawing a Product sets status to Withdrawn and hides it from customers immediately. Vendor can re-activate at any time (re-enters Pending Review if images or declarations changed).

Rejected Listings show the PA rejection reason. Vendor edits and resubmits, creating a new Pending Review entry.

4.3 Order Management

Order list with filters: status (New, Confirmed, Packed, Dispatched, Delivered, Cancelled, Returned), date range, city. Order detail shows customer name (masked to first name and last initial), delivery address, items, total, payment method, and delivery partner assigned.

Actions per status: New → Confirm or Cancel (with reason). Confirmed → Mark as Packed. Packed → Mark as Ready for Pickup (platform delivery) or Dispatch (self-delivery). Dispatched → Track delivery; Mark as Delivered.

SLA timer: vendor must confirm New orders within 30 minutes or the order auto-cancels and the customer is refunded. Bulk actions: Confirm all New orders, Export to CSV.

4.4 Earnings & Payouts

Earnings summary: Gross Sales, Platform Commission (configurable per vendor), Net Payout, TDS deducted. Payout schedule: weekly, every Monday for the prior week's settled orders. Payout history shows date, amount, UTR number, bank account credited.

Vendor can dispute a payout by raising a ticket with order IDs; PA reviews within 3 business days. Reports downloadable as PDF and CSV by date range.

4.5 Locations & Operating Hours

Add or remove served cities. Set operating hours per city (open/close time, days of week). Set temporary closure (holiday mode) per city with start and end dates. Products from a vendor are hidden in a city where the vendor has holiday mode active.

4.6 Key Data Entities

Entity	Fields
Product	id, vendor_id, name, category, description, mrp, price, unit, sku, images[], declarations_signed, status, pa_status, submitted_at, approved_at, reject_reason
ProductRevision	id, product_id, changed_fields[], new_values, pa_status, submitted_at
Order	id, vendor_id, customer_id, items[], total, status, city, delivery_partner_id, sla_expires_at, created_at
Payout	id, vendor_id, period_start, period_end, gross, commission, tds, net, utr, paid_at
VendorCity	id, vendor_id, city, hours, is_open, holiday_until

4.7 API Endpoints

Method	Path	Description
GET	/api/vendor/:id/products	List vendor products (filter: status)
POST	/api/vendor/:id/products	Create new product
PATCH	/api/vendor/:id/products/:pid	Update product fields
POST	/api/vendor/:id/products/:pid/submit	Submit product for PA review
POST	/api/vendor/:id/products/:pid/withdraw	Withdraw a live product
GET	/api/vendor/:id/orders	List orders (filter: status, date, city)
PATCH	/api/vendor/:id/orders/:oid/confirm	Confirm a new order

Method	Path	Description
PATCH	/api/vendor/:id/orders/:oid/status	Advance order status
GET	/api/vendor/:id/earnings	Earnings summary and payout history
GET	/api/vendor/:id/cities	List serving cities with hours
PATCH	/api/vendor/:id/cities/:cid	Update hours or holiday mode

4.8 Functional Test Cases

TC	Scenario	Steps	Expected Result
VP-01	Submit product with all 5 declarations and 2 images	Fill product, check all 5 declarations, upload 2 valid images, submit	Product enters PA Content Review queue; vendor sees status Pending Review
VP-02	Submit blocked without declarations	Fill product, upload images, leave declarations unchecked	Submit button disabled; tooltip shows declarations required
VP-03	Submit blocked with 0 images	Fill product, check declarations, submit with no images	Submission blocked; error: minimum 2 images required
VP-04	Edit live product creates revision	Edit price on a Live product, save	Original live listing stays visible; new revision enters Pending Review
VP-05	SLA timer auto-cancel	New order received; vendor takes no action for 30 minutes	Order auto-cancelled; customer refunded; vendor notified
VP-06	Confirm then mark packed	Vendor clicks Confirm then Packed	Order status advances correctly; delivery partner notified
VP-07	Rejected listing resubmission	PA rejects listing; vendor edits and resubmits	New Pending Review entry created; rejection reason cleared
VP-08	Holiday mode hides vendor	Vendor sets holiday mode for a city	Products from this vendor are hidden in that city's marketplace

TC	Scenario	Steps	Expected Result
VP-09	Payout dispute	Vendor raises dispute on payout with order IDs	Ticket created; PA notified; vendor sees ticket status in Earnings tab
VP-10	Plugin install and configure	Vendor installs SMS plugin, enters webhook URL	Plugin active; test ping succeeds; activity log shows first entry

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5. Customer Marketplace Portal

5. Customer Marketplace Portal

The Customer Marketplace is the consumer-facing shopping interface. Customers browse live products from verified vendors, add to cart, pay, track orders, and manage their account. Only PA-approved products (status = Live) are visible.

5.1 Discovery and Browse

The home feed shows personalised product recommendations, category quick-links, featured vendors, and a deals carousel. Search spans product name, description, and vendor name, with filters for category, price range, dietary tags (Vegan, Organic, Gluten-Free, Halal), vendor rating, and delivery availability in the selected city. Sort options: Relevance, Price Low to High, Price High to Low, Newest, Highest Rated. Results are always scoped to the customer's selected delivery city.

Product detail pages show image gallery with zoom, name and pricing (MRP, selling price, discount percentage), description, nutritional info, vendor name and rating, stock status, customer reviews, and an Add to Cart button.

5.2 Cart and Checkout

Cart persists across sessions for logged-in users. Multi-vendor carts are grouped into separate fulfilment packages with per-vendor delivery fees. Cart shows items, quantity, unit price, line totals, subtotal, applied discount, delivery fee, GST, and final total.

Checkout steps: select delivery address, select delivery slot (per-vendor, based on operating hours), apply promo code (validates in real-time), choose payment method (UPI, card, netbanking, COD if vendor-enabled), review order summary, confirm payment. After payment: order confirmation page with order ID and estimated delivery time.

5.3 Order Tracking

Order list shows all past and current orders with status badges. Order detail includes items, vendor, payment summary, delivery address, delivery partner, and live tracking link. Status flow: Order Placed, Confirmed, Packed, Dispatched, Delivered. Push and email notifications at each status change. Cancel allowed until Packed status (reason required; refund immediate). Return request allowed within 24 hours of delivery (photo upload required; vendor and PA notified).

5.4 Account and Profile

Profile: name, email, phone, profile photo. Saved addresses: add, edit, set default, archive. Payment methods: saved cards and UPI for faster checkout. Preferences: dietary filters applied by default. Loyalty points (if PA-enabled): earn on purchases, redeem at checkout. Referral code: share for credit on referred user's first order. Reviews: customer can edit or delete their own review within 7 days.

5.5 Key Data Entities

Entity	Fields
Customer	id, tenant_id, name, email, phone, preferences, loyalty_points, referral_code
Cart	id, customer_id, items[], promo_code, updated_at
Order	id, customer_id, vendor_id, items[], subtotal, discount, delivery_fee, tax, total, payment_method, payment_ref, status, delivery_address_id, slot_id, created_at
Review	id, customer_id, product_id, order_id, stars, text, vendor_response, moderation_status, created_at
Address	id, customer_id, label, line1, line2, city, state, pincode, is_default
PromoCode	id, code, type (pct/flat), value, min_order, max_uses, uses_count, valid_until

5.6 API Endpoints

Method	Path	Description
GET	/api/marketplace/products	Search and browse products (filter: city, category, query)
GET	/api/marketplace/products/:id	Product detail page data
GET	/api/cart	Fetch current cart
POST	/api/cart/items	Add item to cart
PATCH	/api/cart/items/:id	Update item quantity
DELETE	/api/cart/items/:id	Remove item from cart
POST	/api/cart/promo	Apply promo code
POST	/api/orders	Place order (checkout)
GET	/api/orders	Customer order history
GET	/api/orders/:id	Order detail and tracking
POST	/api/orders/:id/cancel	Cancel an order
POST	/api/orders/:id/return	Raise a return request
POST	/api/reviews	Submit a product review

5.7 Functional Test Cases

TC	Scenario	Steps	Expected Result
CM-01	Pending product not visible	Product in Pending Review; customer searches by name	No results returned
CM-02	City filter shows only local products	Customer selects city Mumbai; browses Dairy	Only Mumbai-serving vendor products shown
CM-03	Multi-vendor cart checkout	Customer adds items from two vendors; checks out	Two fulfilment packages; separate delivery fees; one payment
CM-04	Promo code applies discount	Customer enters valid promo code	Discount deducted; total updated
CM-05	Expired promo code rejected	Customer enters expired code	Error: promo code expired; total unchanged

TC	Scenario	Steps	Expected Result
CM-06	Cancel before Packed	Customer cancels Confirmed order	Order cancelled; refund initiated; vendor notified
CM-07	Cancel after Packed blocked	Customer tries to cancel Packed order	Cancel option unavailable; contact support message shown
CM-08	Return request with photo	Customer uploads photo of damaged item within 24h	Return request created; vendor and PA notified
CM-09	Review and vendor response	Customer submits review; vendor responds	Both visible on product page
CM-10	Out-of-stock badge	Vendor marks product Out of Stock	Out of Stock badge shown; Add to Cart button disabled

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6. Delivery Partner Portal

6. Delivery Partner Portal

Delivery Partners are individuals or logistics companies contracted to pick up orders from vendor locations and deliver them to customers. The portal manages onboarding, active delivery runs, earnings, and performance.

6.1 Delivery Partner Types

Type	Description
Individual Rider	Solo gig delivery person; operates in one city
Fleet Operator	Logistics company managing multiple riders
Vendor Self-Delivery	Vendor uses own staff; registered as a delivery partner for tracking

6.2 Onboarding

Required fields: name, phone, city, vehicle type (bicycle / two-wheeler / three-wheeler / four-wheeler), Aadhaar number, driving licence (for motorised vehicles), bank account details (penny-drop verified). Fleet Operators also provide GST certificate and commercial vehicle registrations. PA approves or rejects.

6.3 Active Delivery Features

Order Queue: Available orders are listed sorted by proximity to the partner's current location. Each entry shows vendor name, pickup address, customer delivery address, estimated distance, and estimated pay. Partner has 60 seconds to accept or decline; declined orders are offered to the next available partner. Maximum 3 active simultaneous orders (configurable per city by PA).

Navigation and Tracking: After accepting, step-by-step status updates: Navigate to Vendor, Arrived at Vendor, Picked Up, Navigate to Customer, Arrived at Customer, Delivered. Each update triggers real-time notifications to vendor, customer, and platform.

OTP at Delivery: Customer provides a 4-digit OTP; partner enters it to mark Delivered. If customer is unavailable, partner uploads a photo after a 5-minute wait timer and the system marks the order Delivered with photo proof.

6.4 Earnings and Payouts

Earnings per delivery: base pay + distance bonus + surge multiplier (peak hours or high demand). Payout weekly every Tuesday for prior week's completed deliveries. Earnings dashboard shows deliveries completed, total earned, bonuses, and deductions (e.g. COD cash not deposited). Dispute ticket for any payout discrepancy.

6.5 Performance Dashboard

Metrics shown: acceptance rate, completion rate, on-time delivery rate, average customer rating. Ratings below threshold (acceptance rate below 70% or completion rate below 80%) trigger a warning. Sustained low performance leads to suspension. Streak bonuses for high-volume days.

6.6 Key Data Entities

Entity	Fields
DeliveryPartner	id, name, phone, city, type, vehicle_type, aadhaar, licence, bank_account_id, status, rating
DeliveryRun	id, partner_id, order_id, accepted_at, picked_up_at, delivered_at, distance_km, pay, otp, proof_photo_url
PartnerPayout	id, partner_id, period_start, period_end, runs_count, gross, deductions, net, utr, paid_at

6.7 API Endpoints

Method	Path	Description
GET	/api/delivery/queue	Available orders near partner
POST	/api/delivery/runs/:rid/accept	Accept an order
POST	/api/delivery/runs/:rid/decline	Decline an order
PATCH	/api/delivery/runs/:rid/status	Advance run status
POST	/api/delivery/runs/:rid/otp-verify	Verify customer OTP at delivery
POST	/api/delivery/runs/:rid/proof	Upload proof of delivery photo
GET	/api/delivery/earnings	Earnings summary and payout history

6.8 Functional Test Cases

TC	Scenario	Steps	Expected Result
DP-01	Partner accepts within 60s	Order offered; partner taps Accept within 60s	Order assigned; vendor and customer notified
DP-02	Auto-reassign after 60s	Partner does not respond within 60s	Order offered to next partner; acceptance rate decremented
DP-03	OTP verification at delivery	Customer provides OTP; partner enters it	Order marked Delivered; earnings recorded
DP-04	Proof of delivery on unavailable customer	Customer not home; partner uploads photo after 5-min wait	Order marked Delivered with photo proof
DP-05	Weekly payout	Partner completes 12 deliveries across the week	Payout initiated on Tuesday; UTR appears in history
DP-06	Performance warning at low acceptance	Partner declines 35% of orders over 7 days	Warning notification; acceptance rate shown in red
DP-07	Fleet operator assigns rider	Fleet Operator assigns order to Rider A	Rider A receives assignment; order not visible to other riders

TC	Scenario	Steps	Expected Result
DP-08	COD cash deduction	Partner collects COD cash but does not deposit	Deduction on next payout; partner notified

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7. Chargeback Portal

7. Chargeback Portal

The Chargeback Portal handles post-payment disputes raised by customers with their bank or card issuer (true chargebacks), as well as platform-level refund requests and vendor payment holds. It is accessible to Platform Admins and a dedicated Finance team role.

7.1 Chargeback Types

Type	Initiated By	Typical Cause
True Chargeback	Customer's bank	Customer denies the transaction
Platform Refund	Customer via support ticket	Item not delivered, wrong item, damaged
Vendor Hold	Platform Admin	Fraud suspicion, payout dispute

7.2 Chargeback Lifecycle

A true chargeback begins when Razorpay notifies the platform via webhook. The platform creates a Chargeback record (status: Open) and notifies the Finance team. Stages: Open, Under Review, Evidence Submitted, Won (bank rules for platform; vendor debited for losses) or Lost (bank rules for customer; refund issued from escrow; vendor debited if goods were delivered), Closed.

For Platform Refunds (via support ticket): Open, Under Review, then Approved (refund issued) or Rejected (customer notified with reason).

7.3 Finance Team Features

Chargeback Queue: List with filters for type, status, date range, vendor, and amount band. An urgency flag shows days remaining within the bank-mandated response window (typically 7-21 days depending on card network).

Case Detail: Full order timeline with timestamps, payment record (gateway transaction ID, amount, method), delivery proof (OTP timestamp or proof photo), vendor declaration, and customer claim.

Evidence Submission: Finance compiles order confirmation, delivery proof, and vendor declaration, then submits to Razorpay via API. Submission timestamp recorded; bank response tracked automatically via webhook.

Vendor Account Holds: A vendor's payout is held when outstanding chargeback amount exceeds the threshold (default Rs. 5,000, configurable by PA). Hold lifts automatically when all open chargebacks are resolved. Finance can manually override to release or extend a hold.

Refund Processing: Approved refunds are issued via the original payment method where possible, or to the customer's platform wallet. Refund record stores amount, method, gateway ref, initiated_at, and settled_at.

7.4 Key Data Entities

Entity	Fields
Chargeback	id, order_id, vendor_id, customer_id, type, amount, status, bank_ref, response_due_at, opened_at, closed_at, outcome
ChargebackEvidence	id, chargeback_id, files[], submitted_at, submitted_by, gateway_submission_ref
Refund	id, order_id, chargeback_id, amount, method, gateway_ref, initiated_at, settled_at, status
VendorHold	id, vendor_id, reason, amount_held, held_at, released_at, released_by

7.5 API Endpoints

Method	Path	Description
POST	/api/webhooks/razorpay/chargeback	Receive chargeback event from Razorpay
GET	/api/chargebacks	List all chargebacks (Finance role)
GET	/api/chargebacks/:id	Chargeback case detail
PATCH	/api/chargebacks/:id/status	Update status

Method	Path	Description
POST	/api/chargebacks/:id/evidence	Submit evidence package
POST	/api/refunds	Initiate a platform refund
GET	/api/refunds/:id	Refund status
GET	/api/vendor/:id/holds	Vendor hold status
PATCH	/api/vendor/:id/holds/:hid	Override hold (Finance or PA only)

7.6 Functional Test Cases

TC	Scenario	Steps	Expected Result
CB-01	Chargeback received from gateway	Razorpay webhook fires chargeback event	Record created; Finance notified; status = Open
CB-02	Evidence submitted before deadline	Finance compiles and submits evidence within window	Sent to Razorpay; timestamp recorded; status = Evidence Submitted
CB-03	Chargeback won	Bank rules for platform	Status = Won; vendor debited for chargeback fee only; hold lifted
CB-04	Chargeback lost	Bank rules for customer	Status = Lost; refund issued to customer; vendor debited for order amount
CB-05	Vendor hold triggered	Vendor chargebacks exceed Rs. 5,000	Payout held; vendor sees hold warning in portal
CB-06	Hold lifted on resolution	All open chargebacks for vendor closed	Hold automatically released; next payout proceeds
CB-07	Platform refund approved	Finance approves refund for undelivered item	Refund issued via original payment method; order status = Refunded
CB-08	Platform refund rejected	Finance rejects with reason	Customer notified; order status unchanged; audit log updated
CB-09	Deadline warning	Response window has 2 days remaining	Queue shows red urgency flag; Finance team gets push notification

TC	Scenario	Steps	Expected Result
CB-10	Manual hold override	Finance releases hold before all chargebacks resolved	Hold lifted; reason recorded in audit log; PA notified

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8. Cross-Portal Data Model

8. Cross-Portal Data Model

This section documents the canonical data entities shared across all portals. Backend engineers should treat this as the authoritative schema reference.

8.1 Entity Relationship Overview

Platform owns Tenants. A Tenant can be Individual, Organisation, or Vendor. A Vendor Tenant owns Products and VendorCities. Customers (also Tenants) place Orders referencing Products and Vendors. Orders are fulfilled by Delivery Partners via DeliveryRuns. Payments flow through Razorpay; chargebacks and refunds are tracked separately. Plugins extend functionality at platform or org level.

8.2 Canonical Entity Schemas

Tenant: id, platform_id, name, type (individual/organisation/vendor), plan, gstin, billing_email, status (pending/active/suspended), created_at, upgraded_at

User: id, tenant_id, email, phone, password_hash, role (platform_admin/tenant_admin/org_admin/vendor_admin/vendor_member/customer/viewer), mfa_enabled, mfa_secret, status (active/suspended/blocked), last_active_at, created_at

VendorProfile: id, tenant_id, business_name, business_type, gstin, pan, fssai_licence, kyc_status (pending/verified/rejected), bank_account_id, gmv_mtd, rating, status (setup_pending/active/suspended), created_at

Product: id, vendor_id, name, category, sub_category, description, mrp, price, unit, sku, stock_status (in_stock/out_of_stock/coming_soon), images[], declarations_signed (0-5), status (draft/pending_review/live/rejected/withdrawn), pa_status, submitted_at, approved_at, reject_reason, created_at, updated_at

ProductRevision: id, product_id, changed_fields[], new_values (json), pa_status, submitted_at, approved_at, reject_reason

VendorCity: id, vendor_id, city, hours (open_time, close_time, days_of_week), is_open, holiday_from, holiday_until

Order: id, customer_id, vendor_id, items[] (product_id, quantity, unit_price, line_total), subtotal, discount, delivery_fee, tax, total, payment_method, payment_ref, payment_gateway, status (placed/confirmed/packed/dispatched/delivered/cancelled/returned/refunded), delivery_address_id, delivery_slot_id, sla_expires_at, created_at, updated_at

DeliveryPartner: id, name, phone, city, type (individual/fleet/vendor_self), vehicle_type, aadhaar, licence, bank_account_id, status (pending/active/suspended), rating, created_at

DeliveryRun: id, partner_id, order_id, offered_at, accepted_at, picked_up_at, delivered_at, distance_km, pay, otp, proof_photo_url, status

Plugin: id, name, version, vendor_id (null for platform-owned), category (payments/tax/shipping/auth/notifications/other), capabilities[], permissions[], risk_level (low/medium/high), status (pending/in_review/approved/rejected/published/deprecated), submitted_at, reviewed_at, reviewer_id, reject_reason

OrgPlugin: id, tenant_id, plugin_id, installed_by, config_json, status (active/disabled), installed_at

Chargeback: id, order_id, vendor_id, customer_id, type (chargeback/platform_refund/vendor_hold), amount, status (open/under_review/evidence_submitted/won/lost/closed), bank_ref, response_due_at, outcome, opened_at, closed_at

Refund: id, order_id, chargeback_id, amount, method (original/wallet), gateway_ref, initiated_at, settled_at, status (pending/settled/failed)

AuditLog: id, actor_id, actor_role, action, target_type, target_id, old_value (json), new_value (json), ip_address, timestamp

8.3 Key Relationships

From	Relationship	To
Tenant	has many	Users
Tenant (Vendor)	has one	VendorProfile
VendorProfile	has many	Products
VendorProfile	has many	VendorCities

From	Relationship	To
Product	has many	ProductRevisions
Customer	has many	Orders
Order	belongs to	Vendor, Customer, DeliveryPartner
Order	has many	DeliveryRuns
Order	may have	Chargeback
Chargeback	may have	Refund
Tenant	has many	OrgPlugins
OrgPlugin	belongs to	Plugin

8.4 Index Recommendations

For query performance: `products(vendor_id, status)` for vendor lists and PA review queue; `products(category, status)` for customer browse; `orders(customer_id, created_at)` for order history; `orders(vendor_id, status, created_at)` for vendor management; `orders(status, sla_expires_at)` for SLA timer checks; `delivery_runs(partner_id, status)` for partner active runs; `chargebacks(vendor_id, status)` for hold checks; `chargebacks(response_due_at, status)` for deadline warnings; `audit_log(actor_id, timestamp)` and `audit_log(target_id, target_type)` for audit lookups.

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9. Plugin System Architecture

9. Plugin System Architecture

The platform is plugin-first: core functionality (payments, tax, notifications, shipping) is delivered through plugins rather than hardcoded. This makes the platform extensible and allows third parties to build and publish plugins through a governed marketplace.

9.1 Plugin Categories

Category	Examples	Risk Level
Payments	Razorpay, Stripe, Cashfree	High
Tax	GST engine, TDS calculator	Medium

Category	Examples	Risk Level
Shipping	Delhivery, Shiprocket, self-delivery	Medium
Auth	SAML SSO, OIDC, OTP providers	High
Notifications	SMS (MSG91), Email (SendGrid), Push	Low
Analytics	Google Analytics, Mixpanel	Low
Compliance	FSSAI checker, GST validator	Medium
Other	Custom vendor tools, loyalty engines	Varies

9.2 Plugin Capabilities and Permissions

Every plugin declares the capabilities it needs. Capabilities map to API scopes reviewed by PA before a plugin is published. Key capabilities: `read:orders`, `write:orders`, `read:products`, `read:customers` (masked PII only), `write:webhooks`, `payments:initiate`, `payments:refund`, `auth:sso`, `notifications:send`, `tax:calculate`.

High-risk capabilities (payments:initiate, payments:refund, auth:sso) require PA review and sandbox test results. Medium-risk capabilities are reviewed for correctness. Low-risk capabilities may be auto-approved after code scan.

9.3 Plugin Lifecycle

Development: third-party vendor builds against the Plugin SDK (sandbox API keys, test fixtures, local emulator). Submission: vendor submits via Vendor Portal with name, version, category, description, capabilities list, webhook URL, and sandbox test results. PA Review: PA reviews capabilities and sandbox results, then approves (Published) or rejects with reason. Installation: once Published, available in Plugin Marketplace for Tenant/Org/Vendor Admins to install. Configuration: installer provides credentials (API keys, webhook secrets) stored encrypted at rest. Versioning: new versions go through the same review cycle; installed orgs can upgrade or stay on previous version.

9.4 Plugin SDK Interface

Plugins implement a standard webhook interface. The platform signs every outbound call with HMAC-SHA256 using the plugin's shared secret. Plugin endpoints must verify the signature before processing.

Request format: `POST /plugin-endpoint` with headers `X-Platform-Plugin-Id` and `X-Platform-Signature` (HMAC-SHA256), and JSON body `{ event, payload, tenant_id, timestamp }`.

Response format: `{ status: "ok" | "error", data?, error_message? }`.

9.5 Platform-Owned vs. Vendor-Owned Plugins

Attribute	Platform-Owned	Vendor-Owned
Created by	Platform Admin	Third-party developer
Review required	No (PA creates directly)	Yes (PA must approve)
Available to	All verified vendors	Any tenant once Published
Sandbox requirement	PA discretion	Required for Medium/High risk

9.6 Key Data Entities

Plugin: id, name, version, vendor_id, category, capabilities[], permissions[], risk_level, status, sandbox_results, submitted_at, reviewed_at, reviewer_id, reject_reason

OrgPlugin: id, tenant_id, plugin_id, installed_by, config_json (encrypted), status, installed_at

PluginLog: id, org_plugin_id, event, request_payload (sanitised), response_status, response_ms, error, logged_at

9.7 Functional Test Cases

TC	Scenario	Steps	Expected Result
PL-01	Submit plugin with high-risk capability	Vendor submits plugin requesting payments:initiate	Plugin enters PA review with High risk badge; auto-approve blocked
PL-02	PA approves plugin	PA reviews capabilities and sandbox results, approves	Plugin status = Published; appears in marketplace
PL-03	PA rejects plugin	PA rejects with reason: missing signature verification	Vendor receives rejection; can resubmit corrected version

TC	Scenario	Steps	Expected Result
PL-04	Install plugin as Org Admin	Org Admin installs SMS plugin, enters API key	Plugin active; test ping succeeds; activity log shows first entry
PL-05	Plugin signature verification	Platform sends event; plugin verifies HMAC-SHA256	Plugin processes event; invalid signature returns 401
PL-06	Versioned upgrade	Vendor publishes v2.0 of installed plugin	PA reviews v2.0; on approval, installed orgs see upgrade prompt
PL-07	Plugin disabled	Org Admin disables plugin	Plugin receives no further calls; config preserved for re-enable
PL-08	Platform plugin available to all	PA creates platform-owned payments plugin	All verified vendors see and can install it

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10. Backend Implementation Guide

10. Backend Implementation Guide

This section gives backend engineers a practical roadmap for implementing the platform, covering recommended stack, service decomposition, auth architecture, background jobs, and deployment guidance.

10.1 Recommended Technology Stack

Layer	Choice	Notes
API server	Node.js (Fastify) or Go	Fastify for rapid iteration; Go for high-throughput routes
Database	PostgreSQL 15+	Primary store; JSONB for config and plugin data
Cache	Redis 7	Session tokens, rate limiting, SLA timers, real-time queue
Object storage	AWS S3 or MinIO	Product images, documents, proof-of-delivery photos

Layer	Choice	Notes
Message queue	BullMQ (Redis-backed)	Background jobs: payouts, notifications, webhooks
Search	PostgreSQL full-text (start); Meilisearch (scale)	Product search
Payments	Razorpay	Payments, payouts, penny-drop verification
SMS / Email	MSG91 (SMS), SendGrid (email)	Or via plugin
Push	Firebase Cloud Messaging	Customer and delivery partner apps
Auth	JWT (access 15 min) + refresh token (7 days, Redis)	MFA via TOTP
Infrastructure	Docker Compose (dev); Kubernetes on AWS/GCP (prod)	

10.2 Service Decomposition

For launch, a modular monolith is recommended over microservices: lower operational overhead, easier debugging, faster iteration, with clear module boundaries that allow future extraction.

Core modules: AuthModule (registration, login, JWT, MFA, SSO); TenantModule (CRUD, upgrades, plan/billing); VendorModule (KYC application, documents, bank verification, cities); ProductModule (catalogue, image upload/validation, declarations, PA review queue, revisions); OrderModule (cart, checkout, order state machine, SLA timers); DeliveryModule (partner onboarding, geo-ordered queue, run lifecycle, OTP, proof upload); PaymentModule (Razorpay integration, webhooks, payout scheduling, chargeback ingestion); PluginModule (submission, review, installation, encrypted config, webhook dispatch with HMAC signing, plugin logs); NotificationModule (email, SMS, push, templates); AuditModule (intercepts writes on sensitive entities, exposes read API).

10.3 Authentication Architecture

Access tokens are JWTs (15-minute expiry) containing sub (user ID), tenant_id, role, and permissions[]. Refresh tokens are opaque random strings stored in Redis with 7-day TTL, rotated on each use. MFA is TOTP-based; on login with MFA enabled, a TOTP challenge is required before the access token is returned. Platform Admin accounts enforce MFA at middleware level.

API route middleware stack (in order): verifyJWT (parse and verify token), resolveRole (extract role and permissions), requireRole([...roles]) (gate to allowed roles). Service accounts authenticate via `Authorization: ApiKey` header; API keys are stored as HMAC hashes.

10.4 Critical Background Jobs

Job	Trigger	Action
SLA Timer	Order placed	Redis TTL key for 30 min; on expiry auto-cancel order and initiate refund
Vendor Payout	Monday 02:00 IST	Aggregate settled orders; calculate commission and TDS; initiate Razorpay payout; record UTR
Partner Payout	Tuesday 02:00 IST	Aggregate completed runs; calculate pay and deductions; initiate payout
Chargeback Alert	Daily 08:00 IST	Check chargebacks with response_due_at within 48 hours; push to Finance team
Plugin Webhook Retry	On failed delivery	Exponential backoff (5s, 30s, 5m, 1h, 24h); after 5 failures mark dead; alert Org Admin
Image Processing	On product image upload	Resize to 800x800 px, convert to WebP, validate white background; store variants in S3
Invitation Expiry	Hourly cron	Mark expired invites; prompt Tenant Admin to re-invite

10.5 Image Upload and Validation

Server-side enforcement (not client-side only): validate MIME type (JPEG or PNG), validate file size (max 5 MB), use the `sharp` library to check pixel dimensions (reject below 800x800), analyse edge-pixel average brightness to detect white background, strip EXIF metadata, resize to 800x800, convert to WebP at 85% quality, generate 200x200 thumbnail. Store originals in S3 under `uploads/{vendor_id}/{product_id}/original/`, WebP under `products/{product_id}/`, thumbnails under `products/{product_id}/thumb/`.

10.6 Webhook Security

Outbound plugin webhook calls include `X-Platform-Signature` computed as HMAC-SHA256(shared_secret, timestamp + "." + JSON.stringify(body)) and `X-Platform-Timestamp`. Plugin endpoints must reject calls older than 5 minutes (replay attack prevention) or with invalid signatures. Inbound webhooks from Razorpay are verified using Razorpay's signature verification SDK before any processing.

10.7 Multi-Tenancy Implementation

All data tables include a `tenant_id` column. Every query appends `AND tenant_id = :current_tenant_id` via query-builder middleware (or PostgreSQL Row-Level Security). Platform Admin routes bypass tenant scoping via a dedicated admin connection. Cross-tenant reads (PA viewing all vendors) explicitly use the admin connection.

10.8 Implementation Sequence (MVP)

1. AuthModule and TenantModule (foundation for all others)
2. VendorModule: KYC application and PA approval flow
3. ProductModule: catalogue management and PA content review queue
4. OrderModule: cart, checkout, order lifecycle and SLA timers
5. PaymentModule: Razorpay integration and basic payout
6. DeliveryModule: partner onboarding and run tracking
7. PluginModule: marketplace and installation
8. ChargebackModule: chargeback and refund handling
9. NotificationModule: email, SMS, push
10. AuditModule: full audit trail

Phase 2 post-MVP additions: SSO integration, fleet operator features, loyalty points, personalised product recommendations, advanced analytics dashboards, and native mobile apps for customers and delivery partners.

Section 1.2 – Sales, Digital Marketing & Publicity Modules

1.2.1 Sales Module

Overview

The Sales module enables the platform's internal sales team to manage outbound vendor acquisition, track revenue and commissions, run invite campaigns, and set and monitor team targets. It integrates with the Vendor/Tenant onboarding flow and the Finance module.

Sub-Modules

1.2.1.1 Lead Pipeline

Manage prospective vendors and tenants from first contact through onboarding.

Field	Type	Notes
lead_id	UUID PK	Auto-generated
business_name	string	Prospect business name
contact_name	string	Primary contact
contact_email	string	
contact_phone	string	
category	enum	vendor / tenant / both
stage	enum	new / contacted / demo_scheduled / negotiating / converted / lost
assigned_to	FK → team_member	Sales rep
source	enum	organic / invite_campaign / referral / direct
notes	text	Internal notes
created_at	timestamp	
updated_at	timestamp	

1.2.1.2 Commission & Revenue Tracking

Track commissions earned by sales reps per vendor/tenant onboarded.

Field	Type	Notes
commission_id	UUID PK	
vendor_id	FK → vendors	Onboarded vendor
sales_rep_id	FK → team_member	
gmv_month	decimal	Gross Merchandise Value that month
commission_rate	decimal	% agreed
commission_amount	decimal	Computed: gmv × rate
status	enum	pending / approved / paid
period	date	YYYY-MM
paid_at	timestamp	Nullable

1.2.1.3 Outbound Vendor Invite Campaigns

Create bulk invite campaigns targeting categories of prospects.

Field	Type	Notes
campaign_id	UUID PK	
name	string	Campaign name
channel	enum	email / whatsapp / sms
target_category	string	Vendor category to target
template_id	FK → message_templates	
scheduled_at	timestamp	
status	enum	draft / scheduled / running / completed
sent_count	int	
opened_count	int	
converted_count	int	Leads who converted
created_by	FK → team_member	

1.2.1.4 Sales Team & Targets

Define sales team members and monthly/quarterly targets.

Field	Type	Notes
target_id	UUID PK	
sales_rep_id	FK → team_member	
period	date	YYYY-MM or YYYY-Q1
target_type	enum	leads / conversions / revenue / gmv
target_value	decimal	
achieved_value	decimal	Updated as events occur
status	enum	on_track / at_risk / missed / achieved

API Endpoints – Sales

Method	Endpoint	Description
GET	/api/v1/sales/leads	List leads (filter: stage, assigned_to, source)
POST	/api/v1/sales/leads	Create new lead
PATCH	/api/v1/sales/leads/:id	Update lead stage or details
DELETE	/api/v1/sales/leads/:id	Archive lead
GET	/api/v1/sales/commissions	List commissions (filter: rep, period, status)
POST	/api/v1/sales/commissions	Log commission entry
PATCH	/api/v1/sales/commissions/:id/approve	Approve commission
PATCH	/api/v1/sales/commissions/:id/pay	Mark as paid
GET	/api/v1/sales/campaigns	List invite campaigns
POST	/api/v1/sales/campaigns	Create campaign
POST	/api/v1/sales/campaigns/:id/launch	Launch scheduled campaign
GET	/api/v1/sales/campaigns/:id/stats	Campaign analytics
GET	/api/v1/sales/targets	List targets (filter: rep, period)
POST	/api/v1/sales/targets	Set target

Method	Endpoint	Description
PATCH	/api/v1/sales/targets/:id	Update target achieved value

1.2.1.5 Functional Test Cases – Sales

TC ID	Scenario	Input	Expected Output
TC-S-001	Create lead	Valid payload	201, lead created in stage=new
TC-S-002	Advance lead stage	Stage: contacted→demo_scheduled	200, stage updated, updated_at refreshed
TC-S-003	Convert lead to vendor	Stage set to converted	Vendor onboarding flow triggered
TC-S-004	Log commission	Valid vendor_id + rep + rate	Commission entry created, status=pending
TC-S-005	Approve commission	PATCH /approve	Status becomes approved
TC-S-006	Launch campaign	POST /launch on draft campaign	Emails/SMS queued, status=running
TC-S-007	Campaign stats	GET /stats	Returns sent/opened/converted counts
TC-S-008	Set sales target	Valid target payload	Target created
TC-S-009	Target auto-update	Conversion event fires	achieved_value incremented, status recalculated
TC-S-010	Filter leads by stage	?stage=negotiating	Only negotiating leads returned

1.2.2 Digital Marketing Module

Overview

The Digital Marketing module covers omnichannel campaign management, promotional pricing via promo codes, SEO/listing boosts for vendors, and analytics on reach and conversions. It works alongside the Sales and Vendor modules.

Sub-Modules

1.2.2.1 Campaign Management (Email / SMS / WhatsApp)

Field	Type	Notes
campaign_id	UUID PK	
name	string	
channel	enum	email / sms / whatsapp
audience_segment	enum	all_vendors / all_customers / specific_category / custom_list
template_id	FK → message_templates	
scheduled_at	timestamp	
status	enum	draft / scheduled / running / paused / completed
sent_count	int	
delivered_count	int	
opened_count	int	Email/WhatsApp only
click_count	int	
unsubscribe_count	int	
created_by	FK → team_member	

1.2.2.2 Promo Codes & Discount Engine

Field	Type	Notes
promo_id	UUID PK	
code	string UNIQUE	e.g. SAVE20
discount_type	enum	flat / percent / free_delivery
discount_value	decimal	
min_order_value	decimal	Nullable
max_uses	int	Nullable = unlimited

Field	Type	Notes
uses_per_user	int	Default 1
valid_from	timestamp	
valid_until	timestamp	
applicable_to	enum	all / specific_vendors / specific_categories
vendor_ids	UUID[]	If applicable_to=specific_vendors
status	enum	active / paused / expired
created_by	FK → team_member	

1.2.2.3 SEO & Listing Boost

Field	Type	Notes
boost_id	UUID PK	
vendor_id	FK → vendors	
boost_type	enum	featured / top_search / category_highlight
start_date	date	
end_date	date	
budget	decimal	Nullable (internal allocation)
impressions	int	
clicks	int	
conversions	int	
status	enum	active / scheduled / expired
created_by	FK → team_member	

1.2.2.4 Analytics (CTR, Conversions, Reach)

Aggregate analytics snapshots computed by background jobs.

Field	Type	Notes
analytics_id	UUID PK	
entity_type	enum	campaign / promo / boost / platform

Field	Type	Notes
entity_id	UUID	FK to the entity
period	date	Daily snapshot
impressions	int	
reach	int	Unique users
clicks	int	
ctr	decimal	clicks / impressions
conversions	int	
conversion_rate	decimal	conversions / clicks
revenue_attributed	decimal	

API Endpoints – Digital Marketing

Method	Endpoint	Description
GET	/api/v1/marketing/campaigns	List campaigns
POST	/api/v1/marketing/campaigns	Create campaign
PATCH	/api/v1/marketing/campaigns/:id	Update campaign
POST	/api/v1/marketing/campaigns/:id/launch	Launch campaign
POST	/api/v1/marketing/campaigns/:id/pause	Pause running campaign
GET	/api/v1/marketing/campaigns/:id/stats	Campaign analytics
GET	/api/v1/marketing/promos	List promo codes
POST	/api/v1/marketing/promos	Create promo code
PATCH	/api/v1/marketing/promos/:id	Update promo
POST	/api/v1/marketing/promos/validate	Validate code at checkout
GET	/api/v1/marketing/boosts	List boosts
POST	/api/v1/marketing/boosts	Create listing boost
PATCH	/api/v1/marketing/boosts/:id	Update boost
GET	/api/v1/marketing/analytics	Platform-wide analytics (filter: entity_type, period)

Method	Endpoint	Description
GET	/api/v1/marketing/analytics/:entity_type/:id	Entity-specific analytics

1.2.2.5 Functional Test Cases – Digital Marketing

TC ID	Scenario	Input	Expected Output
TC-DM-001	Create email campaign	Valid payload, channel=email	201, status=draft
TC-DM-002	Launch campaign	POST /launch	Emails queued, status=running
TC-DM-003	Pause campaign	POST /pause	Status=paused, no new sends
TC-DM-004	Create promo code	code=SAVE20, percent, 20%	201, promo active
TC-DM-005	Validate promo code	POST /validate with SAVE20	Returns discount amount
TC-DM-006	Expired promo rejected	valid_until in past	400, code_expired
TC-DM-007	Exceed max_uses	Uses > max_uses	400, promo_exhausted
TC-DM-008	Create listing boost	featured, 14-day period	201, boost scheduled
TC-DM-009	Analytics CTR	GET /analytics?entity_type=campaign	Returns ctr per campaign
TC-DM-010	Analytics conversion	GET with entity_id	Returns conversion_rate

1.2.3 Publicity Module

Overview

The Publicity module enables the platform team to publish press releases, manage brand/media assets, showcase vendor success stories, and schedule social media posts — all from within the Platform Admin portal.

Sub-Modules

1.2.3.1 Press Releases & Announcements

Field	Type	Notes
press_id	UUID PK	
title	string	
body	text	Rich text / markdown
status	enum	draft / review / published / archived
published_at	timestamp	Nullable
author_id	FK → team_member	
tags	string[]	e.g. [funding, feature, partnership]
slug	string UNIQUE	URL slug
thumbnail_url	string	
created_at	timestamp	
updated_at	timestamp	

1.2.3.2 Media Kit & Brand Assets

Field	Type	Notes
asset_id	UUID PK	
name	string	
asset_type	enum	logo / brand_guide / press_photo / video / font
file_url	string	CDN URL
file_size_kb	int	
mime_type	string	
version	string	e.g. v2.1
is_public	boolean	Publicly downloadable
uploaded_by	FK → team_member	
uploaded_at	timestamp	

1.2.3.3 Vendor Success Stories / Case Studies

Field	Type	Notes
story_id	UUID PK	
vendor_id	FK → vendors	
title	string	
body	text	Rich text
metrics	jsonb	e.g. {"revenue_growth":"3x","orders_month":"1200"}
status	enum	draft / approved / published / archived
published_at	timestamp	
created_by	FK → team_member	
thumbnail_url	string	
slug	string UNIQUE	

1.2.3.4 Social Media Post Scheduler

Field	Type	Notes
post_id	UUID PK	
platform	enum	instagram / facebook / twitter / linkedin / youtube
content	text	Post body
media_urls	string[]	Attached images/video
scheduled_at	timestamp	
status	enum	draft / scheduled / published / failed
published_at	timestamp	Nullable
linked_entity_type	enum	press_release / story / campaign / promo / none
linked_entity_id	UUID	Nullable
created_by	FK → team_member	

Field	Type	Notes
external_post_id	string	ID returned by social API

API Endpoints – Publicity

Method	Endpoint	Description
GET	/api/v1/publicity/press	List press releases
POST	/api/v1/publicity/press	Create press release
PATCH	/api/v1/publicity/press/:id	Update / change status
POST	/api/v1/publicity/press/:id/publish	Publish press release
GET	/api/v1/publicity/assets	List media kit assets
POST	/api/v1/publicity/assets	Upload brand asset
DELETE	/api/v1/publicity/assets/:id	Delete asset
GET	/api/v1/publicity/stories	List vendor success stories
POST	/api/v1/publicity/stories	Create story
PATCH	/api/v1/publicity/stories/:id	Update / approve / publish
GET	/api/v1/publicity/social	List scheduled posts
POST	/api/v1/publicity/social	Schedule a social media post
PATCH	/api/v1/publicity/social/:id	Edit scheduled post
DELETE	/api/v1/publicity/social/:id	Cancel scheduled post
GET	/api/v1/publicity/social/:id/status	Check publish status

1.2.3.5 Functional Test Cases – Publicity

TC ID	Scenario	Input	Expected Output
TC-P-001	Create press release	Valid payload, status=draft	201, draft created
TC-P-002	Publish press release	POST /publish	status=published, published_at set

TC ID	Scenario	Input	Expected Output
TC-P-003	Upload brand asset	POST /assets, logo PNG	201, asset_id, cdn_url returned
TC-P-004	List public assets	GET /assets?public=true	Only is_public=true assets
TC-P-005	Create success story	vendor_id + body + metrics	201, status=draft
TC-P-006	Approve story	PATCH status=approved	Story marked approved
TC-P-007	Publish story	PATCH status=published	Published, slug accessible publicly
TC-P-008	Schedule social post	platform=instagram, future scheduled_at	201, status=scheduled
TC-P-009	Cancel scheduled post	DELETE /social/:id	200, status=cancelled
TC-P-010	Post publish status	GET /social/:id/status	Returns external_post_id or failed reason

1.2.4 Platform-Native Marketing Techniques (App & Portal)

This section extends Section 1.2.2 with platform-native digital marketing strategies suited to a marketplace SaaS like FairShare. These are high-intent, automated, and portal-first — executed inside the platform or through integrations with mobile ad networks and analytics tools.

1.2.4.1 App Store Optimization (ASO)

FairShare's marketplace app (Android / iOS) is optimized for organic discovery through the app stores.

Field	Type	Notes
aso_id	UUID PK	
platform	enum	android / ios
keyword	string	Target keyword
volume	int	Monthly search volume estimate

Field	Type	Notes
rank	int	Current rank for this keyword
screenshot_set_id	UUID	FK → media assets (localized screenshots)
metadata_locale	string	e.g. en-IN, hi-IN
install_rate	decimal	Organic installs from this keyword (7d)
updated_at	timestamp	Last updated

Key actions: keyword research, A/B screenshot testing, localized metadata per region (Hindi, Tamil, Telugu), ratings & review management.

1.2.4.2 In-App & Push Notification Marketing

Behavior-triggered, personalized messages sent to active users on mobile.

Field	Type	Notes
notification_id	UUID PK	
trigger_event	enum	cart_abandon / first_visit / inactivity_7d / new_vendor_near / promo_expiry
channel	enum	push / in_app_banner / in_app_modal
segment	enum	all / new_users / returning / high_value / at_risk
template_id	FK → message_templates	
personalization	jsonb	Dynamic vars e.g. {"vendor_name": "FastShip"}
sent_count	int	
open_rate	decimal	
ctr	decimal	
conversion_rate	decimal	
scheduled_at	timestamp	Or event-triggered
status	enum	draft / active / paused / completed

1.2.4.3 Performance & Paid App Ads

Paid acquisition across mobile ad networks and social portals.

Field	Type	Notes
ad_id	UUID PK	
network	enum	meta / google_uac / linkedin / tiktok / amazon / flipkart / gpay / paytm
campaign_name	string	
objective	enum	app_install / lead_gen / retargeting / awareness
budget_daily	decimal	₹
spend_to_date	decimal	
impressions	int	
clicks	int	
installs	int	App installs attributed
cpi	decimal	Cost per install
roas	decimal	Return on ad spend
status	enum	active / paused / ended
start_date	date	
end_date	date	

1.2.4.4 In-Portal Programmatic Advertising

Real-time bidding (RTB) ad slots inside the FairShare portal itself, enabling vendors to pay for placement.

Field	Type	Notes
slot_id	UUID PK	
slot_type	enum	banner / search_top / category_inject / homepage_hero
vendor_id	FK → vendors	Winning bidder
bid_price	decimal	CPM or CPC

Field	Type	Notes
impressions_served	int	
clicks	int	
ctr	decimal	
start_at	timestamp	
end_at	timestamp	
status	enum	active / expired / scheduled

Programmatic slots are auctioned via a simple second-price auction; bids are submitted through the Vendor portal and reviewed by the Platform Admin before going live.

1.2.4.5 Deep Linking & Universal Links

Directs users from any channel (email, social ad, web search, QR) straight into a specific in-app screen, removing friction.

Field	Type	Notes
deeplink_id	UUID PK	
destination	enum	product_page / vendor_page / promo / checkout / category
destination_id	UUID	FK to the target entity
short_url	string	e.g. fairshare.app/go/abc123
source	string	utm_source value
medium	string	utm_medium value
campaign	string	utm_campaign value
clicks	int	
app_opens	int	Users who opened app via this link
conversions	int	Completed checkout
created_by	FK → team_member	
created_at	timestamp	

1.2.4.6 Affiliate & Influencer Portal Partnerships

Partner creators and affiliate networks promote FairShare using unique referral links with tracked conversion codes.

Field	Type	Notes
partner_id	UUID PK	
name	string	Creator / network name
type	enum	affiliate / influencer / partner_network
platform	string	e.g. YouTube, Instagram, cashback_network
referral_code	string UNIQUE	
commission_type	enum	cpa / cpc / revenue_share
commission_value	decimal	Fixed ₹ or %
clicks	int	
signups	int	
conversions	int	
revenue_attributed	decimal	
status	enum	active / paused / terminated
contract_url	string	Link to signed agreement

1.2.4.7 Full-Funnel Behavioral Tracking (RACE Framework)

All user journeys mapped from first touchpoint to retention using the RACE framework (Reach → Act → Convert → Engage).

Stage	Metric	Source
Reach	App installs, Ad impressions, ASO rank	Ad networks, App Store API
Act	App opens, Browse sessions, Search queries	In-app analytics
Convert	Add-to-cart, Checkout start, Order placed	Platform events
Engage	Repeat orders, Review submitted, Referral sent	CRM, notification opens

Data pipelines aggregate events into the `analytics` table (Section 1.2.2.4) with `entity_type = 'funnel'`. A RACE dashboard in the Platform Admin portal visualizes the full funnel, drop-off rates per stage, and cohort retention curves.

API Endpoints – Platform-Native Marketing

Method	Endpoint	Description
GET	/api/v1/marketing/aso	List ASO keyword rankings
POST	/api/v1/marketing/aso	Add/update keyword tracking
GET	/api/v1/marketing/notifications	List push/in-app notification campaigns
POST	/api/v1/marketing/notifications	Create notification trigger
PATCH	/api/v1/marketing/notifications/:id/pause	Pause notification
GET	/api/v1/marketing/ads	List paid ad campaigns
POST	/api/v1/marketing/ads	Create ad campaign (linked to external network)
PATCH	/api/v1/marketing/ads/:id	Update budget / status
GET	/api/v1/marketing/programmatic/slots	List ad slots
POST	/api/v1/marketing/programmatic/slots	Create programmatic slot
GET	/api/v1/marketing/deeplinks	List deep links
POST	/api/v1/marketing/deeplinks	Generate deep link
GET	/api/v1/marketing/deeplinks/:id/stats	Click and conversion stats
GET	/api/v1/marketing/affiliates	List affiliate/influencer partners
POST	/api/v1/marketing/affiliates	Create affiliate partner
GET	/api/v1/marketing/affiliates/:id/stats	Partner attribution stats
GET	/api/v1/marketing/funnel	RACE funnel metrics (filter: period, segment)

1.2.4.8 Functional Test Cases – Platform-Native Marketing

TC ID	Scenario	Input	Expected Output
TC-PN-001	Add ASO keyword	keyword=marketplace, platform=android	201, tracking started
TC-PN-002	Trigger push notification	trigger=cart_abandon, segment=returning	Notification queued for all returning users with abandoned carts
TC-PN-003	Push ctr tracked	User opens notification	open_rate and ctr incremented
TC-PN-004	Create paid ad	network=meta, objective=app_install	201, ad campaign created in draft
TC-PN-005	Ad ROAS calculated	spend=₹10K, revenue=₹40K	roas=4.0 returned
TC-PN-006	Create programmatic slot	slot_type=homepage_hero, bid=₹200 CPM	201, slot pending review
TC-PN-007	Generate deep link	destination=promo, promo_id=PR-001	Short URL returned, click tracking active
TC-PN-008	Deep link conversion	User opens app via link, completes order	conversions incremented
TC-PN-009	Create affiliate	name=Creator, referral_code=BLOG10	201, code unique, partner active
TC-PN-010	Affiliate conversion	Order with referral_code=BLOG10	revenue_attributed updated, commission logged
TC-PN-011	RACE funnel pull	GET /funnel?period=2026-10	Returns reach/act/convert/engage metrics

TC ID	Scenario	Input	Expected Output
TC-PN-012	Duplicate referral code	POST with existing code	409, code_conflict